

Lim, Hyeon Jun

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EDUCATION

Master of Science in Electrical and Electronic Engineering

Feb. 2018

Yonsei University, Seoul, South Korea

- Thesis Title: “State Estimation using Sampled-data Fuzzy Observer and its Application to AHRS”

Bachelor of Science in Electrical and Electronic Engineering

Feb. 2016

Yonsei University, Seoul, South Korea

- National Science and Engineering Scholarship (awarded by the Government of South Korea)

PROFESSIONAL EXPERIENCE SUMMARY

Senior Research Engineer with 7+ years of experience in robotics and autonomous driving, specializing in Reinforcement Learning (RL), Transformer-based End-to-End (E2E) planning, and high-reliability decision-making systems. Demonstrated expertise in architecting closed-loop simulation environments, developing advanced LiDAR-based perception and 3D reconstruction systems, and integrating real-vehicle software for production-grade autonomous platforms. Built and managed large-scale autonomous-driving datasets exceeding 1M labeled objects, and designed LiDAR simulation frameworks supporting sensor-technology development, HD mapping, and high-fidelity 3D scanning. Strong theoretical foundation in T-S fuzzy-based robust control, complemented by hands-on experience delivering scalable, safety-critical algorithms and infrastructure across both research and industrial environments.

EMPLOYMENT

Hyundai Motor Company, Seoul, South Korea

- Senior Research Engineer, Autonomous Driving Development Jan. 2025 – Present
- Research Engineer, Autonomous Driving Development Mar. 2022 – Dec. 2024

Korea Institute of Industrial Technology, Ansan, South Korea

- Researcher, Robotics R&BD Group Jun. 2018 – Jun. 2021

KEY PROJECTS

Project: Development of a Unified Closed-Loop Simulation Framework for Reinforcement Learning-Based Autonomous Driving

Aug.2025 - Present

- **Architected a high-fidelity closed-loop autonomous-driving simulation stack**—combining 3DGS-based photorealistic environments, rule-driven reward modeling, and World-Model-powered synthetic data generation—to enable reactive, long-tail scenario evaluation and robust Sim-to-Real generalization.
- **Optimized and validated real-world driving policies** by integrating Online/Offline RL paradigms, designing specialized neural layers for transfer, and successfully deploying the resulting agents on physical vehicles, achieving major gains in success rate, safety, and collision avoidance.

Project: Development of End-to-End Autonomous Planning System

Jul. 2024 – Mar. 2026

- Led the **end-to-end driving software development**, embedded deployment, and real-world validation of a Transformer-based **autonomous-driving** planner—building a unified training/inference framework, enabling **first-ever on-vehicle control**, architecting closed-loop evaluation and long-tail robustness analysis with high-fidelity synthetic data, and accelerating iteration through custom real-time visualization and debugging tools used across **diverse real-vehicle tests**.

Project: Development of Lv.3 and Lv.4 Autonomous Driving Perception Systems (HDP & Robo-Services)

Mar. 2022 – Jun. 2024

- Led the software architecture, development, and deployment of the company's **LiDAR perception stack**—establishing the first systematic **sensor-evaluation** framework, benchmarking 10+ vendors, developing core parsing, filtering, fault-detection, clustering, localization, and calibration technologies (including online optical-flow calibration), optimizing embedded performance with **CUDA**, and validating the full system across Lv.3/Lv.4 platforms such as **Genesis G90**, **Kia EV9**, and RoboRide/RoboShuttle/RoboDelivery services.

Project: Development of River Management Technology Using a Drone-Based Platform for River Investigation

Jun. 2019 – Jun. 2021

- Developed a high-fidelity **LiDAR simulation** and analysis framework to derive optimal hardware specifications, validate **dual-wavelength bathymetric sensing performance**, and translate point-cloud-based error modeling into actionable embedded-system requirements for high-accuracy river and subsurface mapping.

Project: Development of a Standardized Open Database and Evaluation System for Intelligent Vehicle Perception

Jun. 2018 – May 2020

- Built **South Korea's first large-scale, high-precision LiDAR dataset** for **autonomous driving**—constructing a million-label 128-channel corpus with automated annotation (90% time reduction), long-tail extreme-condition captures, cm-level HD mapping with custom tooling (LAS converter, point-wise sync, SOR filtering, rolling-shutter correction), and a full-stack calibrated sensor-logging platform.
- Advanced 3D perception fidelity and validation rigor by boosting Z-axis accuracy from 10 cm to 1 cm via global ground-fitting, performing comprehensive multi-sensor calibration, and benchmarking SOTA **deep-learning models** to confirm real-world applicability and robustness of the dataset.

Project: Development of a 3D Vehicle Shape Generation System for Automated Car Wash

Jan. 2019 – Dec. 2020

- Developed a fast, sub-2 cm-accuracy **3D vehicle-profiling system** for unmanned car-wash automation, integrating custom **multi-LiDAR sensing**, fused point-cloud reconstruction, and **real-time TI-embedded processing** to ensure reliable geometry extraction and collision-free operation.

PUBLICATIONS

- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, "Relaxed Stabilization Conditions for Discrete T-S Fuzzy Models under Imperfect Premise Matching," Journal of Korean Institute of Intelligent Systems [KCI, IF 0.47], 2017.
- Hyeon Jun Lim, Chang Gil Lee, Jin Bae Park, and Young Hoon Joo*, "Design of Output-feedback Controller for Discrete T-S Fuzzy Models under Imperfect Premise Matching," Journal of Information and Control [KCI], 2016.
- Hyeon Jun Lim, Seung Ho Baeg*, "Calibration Technique between 2D LiDAR Sensor and Linear Encoder using Dihedral Angle," Journal of Information and Control [KCI], 2020.
- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, "Fuzzy Observer Design and Its Experimental Validation Using the Fuzzy Lyapunov Function," 17th International Conference on Control, Automation and Systems (ICCAS) [International Conference], 2017.
- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, "Fuzzy Lyapunov Function-based H_∞ Fuzzy Filter Design for Sampled-data Systems under Imperfect Premise Matching," Proceedings of the SICE Annual Conference [International Conference], 2017.
- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, "The Stabilization Condition of Continuous Affine Fuzzy Systems Under Imperfect Premise Matching," International Conference on Control, Automation and Systems (ICCAS) [International Conference], 2016.
- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, "The Fuzzy Lyapunov Function-based Relaxed Stabilization of Output-feedback Fuzzy Control Systems Under Imperfect Premise Matching," International Conference on Fuzzy Theory and It's Applications (iFuzzy) [International Conference], 2016.

- Hyeon Jun Lim, Seung Ho Baeg*, “Improving Global Mapping Accuracy of Point Cloud Data using SOR Filter,” 16th Korea Robotics Society Annual Conference (KROC) [Domestic Conference], 2021.
- Hyeon Jun Lim, Seung Ho Baeg*, “Time Compensation Technique of Scanning LiDAR Sensor using Hermite Interpolation,” KIIS Autumn Conference [Domestic Conference], 2020.
- Hyeon Jun Lim, Seung Ho Baeg, Jae Hak Kim, Jae Sung Jeong, and Chang Seong Kim, “Analysis of Design Parameters and Performance Simulation of LiDAR Sensors for River Measurement,” KSCE CONVENTION [Domestic Conference], 2020.
- Seung Ho Baeg, Beom Jin Hong, Su Hyun Na, and Hyeon Jun Lim, “Korean Dataset Construction System for Autonomous Driving Technology Development,” KSAE Annual Conference [Domestic Conference], 2018.
- Hyeon Jun Lim, Jin Bae Park, Taek Kyu Song, and Young Hoon Joo*, “Relaxed Stabilization Conditions for Affine T-S Fuzzy Systems Using Extended Fuzzy Lyapunov Function: Guaranteed Cost Approach,” KIEE Summer Conference [Domestic Conference], 2017.
- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, “Design of Relaxed Output-feedback Controller for T-S Fuzzy Models under Imperfect Premise Matching,” ICROS Annual Conference [Domestic Conference], 2017.
- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, “Design of Output-feedback Controller for T-S Fuzzy Models using Fuzzy Lyapunov Function: Guaranteed Cost Approach,” KISS Spring Conference [Domestic Conference], 2017.
- Hyeon Jun Lim, Jin Bae Park, and Young Hoon Joo*, “Relaxed Stabilization Conditions for Discrete T-S Fuzzy Models under Imperfect Premise Matching,” KIIS Autumn Conference [Domestic Conference], 2016.
- Hyeon Jun Lim, Han Sol Kim, Jin Bae Park, Myoung Hwan Tak, and Young Hoon Joo*, “Stabilization of T-S Fuzzy Systems Using Line-integral Fuzzy Lyapunov Function,” KISS Spring Conference [Domestic Conference], 2016.

PATENTS

- Hyeon Jun Lim, [Co-inventors], “Apparatus For Controlling Vehicle and Method Thereof,” Korean Patent Application, KR 10-2025-0044855, Apr. 7, 2025. (Also filed in US 19/391,095, DE 102025146420.0, and CN 202511797806.2)
- Hyeon Jun Lim, [Co-inventors], “System and Method For Adaptive Calibration For Autonomous Driving,” Korean Patent Application, KR 10-2025-0042965, Apr. 2, 2025. (Also filed in US 19/409,994 and CN 202511849891.2)
- Hyeon Jun Lim, [Co-inventors], “System and Method For Calibrating 3D Sensor Coordinate Based On Scene Flow Network,” Korean Patent Application, KR 10-2025-0042966, Apr. 2, 2025. (Also filed in US 19/406,040 and CN 202511849897.X)
- Hyeon Jun Lim, [Co-inventors], “System For Processing Data Required For Object Detection Based On Artificial Intelligence and Multiple Autonomous Driving Sensors and Method Implementing The Same,” Korean Patent Application, KR 10-2025-0090242, Jul. 4, 2025. (Also filed in US 19/415,494 and CN 202511840140.4)
- Hyeon Jun Lim, [Co-inventors], “3D shape data acquisition method for automatic car wash equipment and vehicle shape data acquisition system,” Korean Patent (Registered), KR 10-2504548-B1, 2023.

SOFTWARE COPYRIGHTS

- Hyeon Jun Lim, Seung Ho Baeg, “Laser Beam Characteristics Evaluation Program,” Software Copyright, Korea Copyright Commission, C-2021-008734, Feb. 2021.
- Hyeon Jun Lim, Seung Ho Baeg, “LiDAR Sensor Data Viewer Program,” Software Copyright, Korea Copyright Commission, C-2020-052146, Dec. 2020.
- Seung Ho Baeg, Hyeon Jun Lim, “Nov Atel GNSS Data Reception Firmware for TI ARM Processor,” Software Copyright, Korea Copyright Commission, C-2020-052144, Dec. 2020.
- Seung Ho Baeg, Hyeon Jun Lim, “Range Data Acquisition Firmware for DSP Processor,” Software Copyright, Korea Copyright Commission, C-2020-052143, Dec. 2020.
- Seung Ho Baeg, Hyeon Jun Lim, “PRU Firmware for Motor Speed Control and Rotation Angle Detection on TI DSP,” Software Copyright, Korea Copyright Commission, C-2020-052142, Dec. 2020.
- Seung Ho Baeg, Hyeon Jun Lim, “CAN and Serial Communication API Driver for TI DSP,” Software Copyright, Korea Copyright Commission, C-2020-052141, Dec. 2020.
- Seung Ho Baeg, Hyeon Jun Lim, “Epson IMU Data Reception Firmware for PRU Processor,” Software Copyright, Korea Copyright Commission, C-2020-052140, Dec. 2020.
- Hyeon Jun Lim, Seung Ho Baeg, “LAS File Conversion Program,” Software Copyright, Korea Copyright Commission, C-2020-052137, Dec. 2020.